Mathematics Extended Part Module 1 (Calculus and Statistics) Mock

Section A (Short Questions) · Section B (Long Questions)

150 minutes (2.5 hours)

Total marks：100

FOR REVISION

For revision and self-assessment only. This is not an official examination paper.

Difficulty rating：**4.0** / 5.0

INSTRUCTIONS

1. This paper consists of Section A and Section B. Answer ALL questions.
2. Unless otherwise specified, all working must be clearly shown.
3. Unless otherwise specified, numerical answers should be either exact or correct to 4 decimal places.
4. The diagrams in this paper are not necessarily drawn to scale.
5. A standard normal distribution table is printed on the last page of this paper.

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* *Independent, unofficial mock paper aligned to the DSE syllabus. Not affiliated with any official examination authority. For revision only.*

Section A (50 marks)

1. Let X be a discrete random variable with probability function $P(X = x) = k(c - x)$, where $x = 1, 2, 3, 4, 5$, and k and c are constants. It is given that $E(X) = \frac{7}{3}$.

(a) Find k and c.

(b) Let $\bar{X}$ be the mean of 56 independent random observations of X. Find $\mathrm{Var}(\bar{X})$ and $P(\bar{X} > 2.5)$.

(7 marks)

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2. Let Y follow a binomial distribution with 6 trials and probability of success p in each trial, where \$0 (a) Find p. (b) Find $P(Y > E(Y))$. (5 marks)

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3. The distance that a brand of electric scooter can travel on a full charge follows a probability distribution with mean μ km. A sample of 64 scooters is randomly selected from a large batch of scooters of this brand, and their distances are denoted by $x_1, x_2, \ldots, x_{64}$ (in km). It is given that $\sum_{i=1}^{64} x_i = 1632$ and $\sum_{i=1}^{64} x_i^2 = 41978.88$. The sample standard deviation is $\sqrt{\frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2\right)}$, where n is the sample size and $\bar{x}$ is the sample mean.

(a) Find the sample mean and the sample standard deviation.

(b)(i) Someone claims that the 95% confidence interval for μ constructed using this sample mean and sample standard deviation does not contain 25. Do you agree? Explain your answer.

(ii) Another sample of m scooters is randomly selected from the large batch, and its sample mean and sample standard deviation are the same as those in (a). If the 95% confidence interval for μ constructed using this sample does not contain 25, find the least value of m.

(7 marks)

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4. Let A and B be two events. It is given that $P(A \mid B) = 0.3$, $P(A \cap B) = 0.12$ and $P(A' \cap B') = 0.4$.

(a) Find $P(B)$ and $P(A)$.

(b) Are A and B independent? Explain your answer.

(c) Let C be an event such that A and C are independent and $P(A \cup C) = 0.66$. Find $P(C)$.

(6 marks)

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5. For all $x \in \mathbf{R}$, define $f(x) = e^{2x} - 6e^x + 4x$.

(a) Prove that $f'(x) = 2(e^x - 1)(e^x - 2)$.

(b) Someone claims that $f(x)$ increases as x increases for all $x > 0$. Do you agree? Explain your answer.

(5 marks)

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6. (a) Let k be a positive constant. Expand k^x in ascending powers of x as far as the term in x^2, giving the answer in terms of $\ln k$.

(b)(i) It is given that there is no x^2 term in the expansion of $(1+x)^4k^x$, and the coefficient of x is positive.

Find the exact value of k.

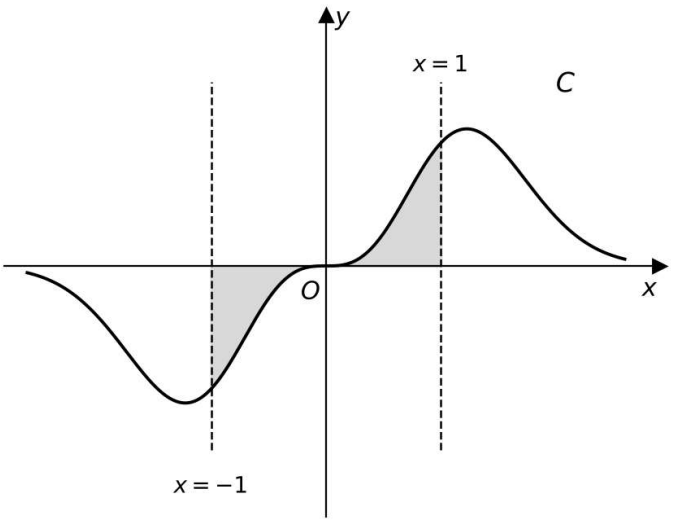
(ii) Using $(1+x)^5k^x = (1+x)\left[(1+x)^4k^x\right]$, find the coefficient of x^2 in the expansion of $(1+x)^5k^x$.

(6 marks)

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7.



Consider the curve C: $y = x^3e^{-x^2}$ (see the figure).

(a) By considering $\frac{d}{dx}\left[(x^2+1)e^{-x^2}\right]$, find $\int x^3e^{-x^2}\,dx$.

(b) Find the total area of the regions bounded by C, the x-axis, the line $x = -1$ and the line $x = 1$.

(6 marks)

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8. Consider the curve Γ: $y = ax + b + \frac{8}{(x+1)^2}$, where $x > -1$, and a and b are constants. It is given that Γ has a turning point at $x = 1$ and Γ passes through the point $(1, 5)$.

(a) Find a and b.

(b)(i) Does Γ have a maximum point? Explain your answer.

(ii) Find the equation of the tangent to Γ at $x = 3$.

(8 marks)

Section B (50 marks)

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9. The number of meteors that can be observed per hour on a clear night at a stargazing campsite follows a Poisson distribution with mean λ. It is given that the probability of observing exactly 2 meteors in a certain hour is 1.2 times the probability of observing exactly 1 meteor in that hour.

(a) Find λ.

(b) Find the probability that at least 2 meteors are observed in a certain 20-minute period.

(c)(i) A stargazer observes meteors for 3 consecutive hours starting from 9 p.m. on clear nights. A night is called *fruitful* if at least 1 meteor is observed in each of these 3 hours. Assume that the numbers of meteors observed on different nights are independent.
Find the probability that the stargazer observes a total of 4 meteors in the 3 hours of a certain night.

(ii) Given that the stargazer observes a total of 4 meteors in the 3 hours of a certain night, find the probability that the night is *fruitful*.

(iii) Let P_n be the probability that exactly $(n-1)$ of n clear nights are *fruitful*, where n is a positive integer. Someone claims that P_n is greatest when $n = 4$. Do you agree? Explain your answer.

(12 marks)

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10. The usage time T hours of a brand of mobile phone, from fully charged to automatic shutdown, follows a normal distribution with mean μ hours and standard deviation σ hours. It is given that 11.51% of the phones of this brand have T greater than 26 and 2.28% of the phones have T less than 18. A phone is called *durable* if $T > 25$, and is called *defective* if $T < 20$.

(a) Find μ and σ.

(b) Given that a phone of this brand is not *defective*, find the probability that it is *durable*.

(c)(i) An inspector randomly selects phones of this brand one by one from a large batch for inspection, until a *defective* phone is selected or 8 phones have been inspected. Find the probability that the first phone inspected is *defective*.

(ii) Find the probability that the inspector inspects 8 phones.

(iii1) Given that the inspector inspects 8 phones, does the expected number of *durable* phones among the 8 phones inspected exceed 1.8? Explain your answer.

(iii2) Given that the inspector inspects 8 phones, find the probability that exactly 2 of the 8 phones inspected are *durable*.

(14 marks)

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11. A ski resort makes snow with snow machines in the evening. Let t ($0 \leq t \leq 4$) be the number of hours elapsed since 6 p.m. During this period, snow is produced at a rate of $8te^{\frac{t^2}{16}}$ m³ per hour, and snow melts at a rate of $10\sqrt{1+e^{\frac{t}{2}}}$ m³ per hour.

(a) Use the trapezoidal rule with 4 sub-intervals to estimate $\int_0^4 \sqrt{1+e^{\frac{t}{2}}}\,dt$.

(b) Using integration by substitution, evaluate $\int_0^4 8te^{\frac{t^2}{16}}\,dt$.

(c)(i) Using (a) and (b), estimate the change in the amount of snow at the ski resort from $t = 0$ to $t = 4$.

(ii) From $t = 0$ to $t = 4$, does the amount of snow at the ski resort increase by more than 29 m³? Explain your answer.

(13 marks)

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12. A fish farm raises fish in a pond. Let t ($t \geq 0$) be the number of months elapsed since the fry were released. It is given that the number of fish in the pond is $5000e^{-\frac{t}{10}}$ and the average mass of the fish is $3\left(1-e^{-\frac{t}{5}}\right)$ kg. Let W (in thousand kg) be the total mass of the fish in the pond.

(a) Find $\frac{dW}{dt}$.

(b) Find the exact value of t such that W attains its greatest value.

(c)(i) Find the greatest value of W.

(ii) Someone claims that in the first 13 months, W is greater than half of its greatest value for more than 11 months. Do you agree? Explain your answer.

(11 marks)

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END OF PAPER

Standard Normal Distribution Table

Each entry is the area under the standard normal curve from 0 to z, i.e. $P(0 \le Z \le z)$. For negative z, use symmetry: $P(-z \le Z \le 0) = P(0 \le Z \le z)$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517

0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

For $z \geq 3.50$, the area may be taken as 0.4998 or above.

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